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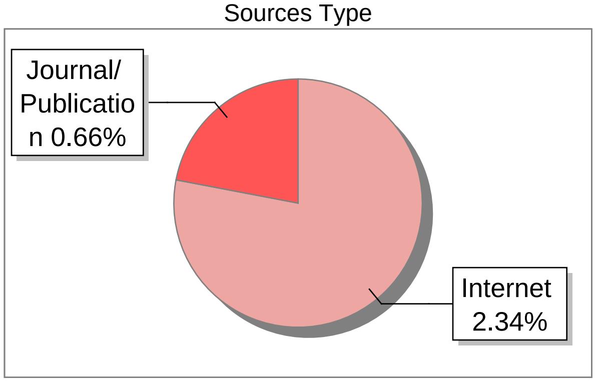

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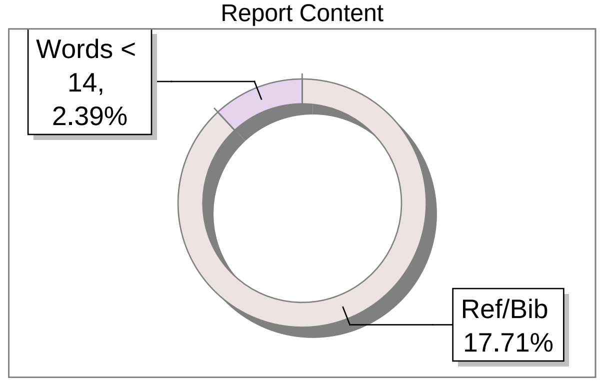

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Restoring Trust on E-Commerce Platforms: Fake Review Detection Using Social Network Analysis and Gradient Boosting

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***Abstract*—Fake reviews negatively affect customer trust and distort the reputation of products on online marketplaces. This paper presents a simple hybrid approach that combines Social Network Analysis (SNA), TF–IDF text features, and behavioural indicators such as helpfulness ratio to detect misleading reviews. Using an Amazon product review dataset, we construct a reviewer–product graph, compute degree centrality, extract TF–IDF features, and train multiple classifiers. Gradient Boosting achieved the strongest performance for the minority fake-review class. The model was also integrated into a lightweight dashboard for real-time testing and visual interpretation.**

***Index Terms*—Fake review detection, e-commerce, social network analysis, TF–IDF, Gradient Boosting, machine learning.**

I. Introduction

Online reviews strongly influence the decisions customers make while shopping on e-commerce platforms. People often depend on star ratings and written feedback, while sellers rely on positive reviews to increase visibility. Because reviews impact product ranking and user trust, the presence of misleading or fake reviews can create serious problems.

Various kinds of fake or manipulated reviews exist, including paid promotional content, duplicate text, review bombing, and organised groups attempting to influence product ratings [1]. Since these reviews appear in large volumes, manual inspection is impractical. Automated approaches are therefore essential.

Social Network Analysis (SNA) helps identify unusual reviewer behaviour by modelling interactions between reviewers and products as a graph [2]. Text analysis techniques like TF–IDF also capture linguistic signals that may reveal unnatural or repetitive writing patterns [3]. This paper introduces a hybrid pipeline that merges SNA, text, and behavioural features to detect fake reviews.

II. Motivation

Fake reviews damage consumer trust and can unfairly promote or demote products. Detecting them helps protect customers and maintains platform reputation. Many past methods focus only on text or simple heuristics; adding network-aware features improves detection and provides interpretable signals about suspicious reviewer hubs.

III. Problem Statement

Given a large set of product reviews from an e-commerce site, design an automated system that flags likely fake reviews. The system should combine structural (SNA), textual (TF–IDF), and behavioural features, handle class imbalance, and be practical to run on consumer-grade hardware.

IV. Objectives

1) Construct a reviewer–product network and compute simple SNA metrics.
2) Extract textual features (TF–IDF) and behavioural indicators (helpfulness ratio).
3) Train and compare classifiers (Logistic Regression, SVM, Random Forest, Gradient Boosting).
4) Select a robust model for the minority fake class and deploy a small demo for real-time checks.
5) Provide visual diagnostics (model comparison and network visualization).

V. Scope

Includes: public Amazon review data, Python-based processing, SNA metrics, TF–IDF text features, basic behavioural metrics, and a demo dashboard. **Excludes:** private user-identifiable data, production-scale deployment, and full recommendation-system integration.

VI. Contributions

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- A compact hybrid pipeline that merges SNA, text, and behavioural features for fake review detection.
- Empirical comparison of four classifiers, highlighting Gradient Boosting for this task.
- A simple demo that illustrates real-time detection and SNA-based visual diagnostics.

VII. Related Work

Early work by Jindal and Liu studied opinion spam by identifying duplicate and near-duplicate reviews. Network-based methods later showed that modelling user–item interactions can expose abnormal behaviour and collusion. More recent surveys highlight hybrid systems that combine linguistic, temporal, and network signals to improve robustness against sophisticated spam and AI-generated reviews. Our approach follows this hybrid tradition but focuses on a compact, explainable feature set suitable for demonstration.

VIII. System Requirements and Specifications

A. Functional Requirements

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- Load and clean review data (text, reviewer ID, product ID, helpful votes, total votes, label).
- Build reviewer–product graph and compute degree centrality.
- Extract TF–IDF text features and compute helpfulness ratio.
- Train classifiers and evaluate using F1-score (fake class).
- Provide simple visual outputs and a demo UI for live testing.

B. Non-functional Requirements

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- Run comfortably on a typical laptop (8GB RAM).
- Modular code structure for easy updates.
- Results should be reproducible via saved preprocessing/model files.

C. Hardware and Software

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- Python 3.8+; packages: pandas, numpy, scikit-learn, networkx, matplotlib, joblib, gradio (optional).
- Standard laptop (Intel i3/Ryzen 3 or better), 10GB free disk.

IX. System Design and Methodology

Figure 1 shows the pipeline: load data, build graph, extract features, train models, evaluate, and demo.

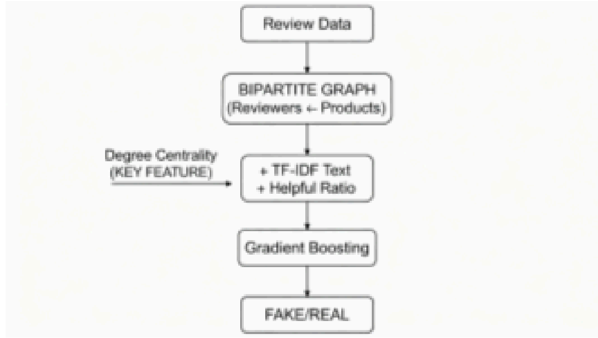

Fig. 1. High-level pipeline: data → features → models → demo.

A. Data Preparation

Load the CSV, drop rows missing critical fields (review text, reviewer ID, product ID), and normalise text by lowercasing and basic tokenisation. The label column `is_fake` identifies known fake reviews if present.

B. Social Network Construction

Create a bipartite graph $G = (R, P, E)$ where R is reviewers and P is products. Add edges for each (reviewer, product) pair. Compute degree centrality for reviewer nodes and attach the score as a numeric feature to each review record. High degree may indicate heavy activity and warrants inspection.

C. Textual and Behavioural Features

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- **TF–IDF:** Fit a TF–IDF vectoriser (max features 1000, English stopwords) on review text.
- **Helpfulness ratio:** helpful_ratio = helpful_votes / (total_votes + ϵ).

Concatenate degree centrality, helpfulness ratio, and TF–IDF vectors into a final feature matrix.

D. Data Splitting and Scaling

Standardise numeric features (degree centrality, helpfulness ratio) with StandardScaler. Split data into training and test sets (70/30) with stratification on the fake label.

E. Model Selection

Evaluate the following classifiers: leftmargin=*

- Logistic Regression (linear baseline)
- Linear SVM
- Random Forest (100 trees)
- Gradient Boosting (e.g., XGBoost or sklearn's HistGradientBoosting)

Use F1-score for the fake class as the primary metric.

X. Implementation

A. Pipeline Overview

Implement the pipeline in Python with modular scripts: leftmargin=*

- `data_prep.py` — load and clean data.
- `features.py` — build graph and compute features.
- `train.py` — train models and save best model with joblib.
- `demo.py` — simple Gradio app for live testing.

B. Model Training and Evaluation

Train each model on the training set, predict on the test set, and compute precision, recall, and F1 for both classes. Because the fake class is usually a minority, consider class weighting or sampling strategies if needed. Save the chosen model, TF–IDF vectoriser, and scaler for deployment.

XI. Dashboard Design

A small Gradio-based interface accepts raw review text, computes features using saved preprocessing objects, and returns the predicted probability and label. The demo also displays static plots: model comparison and a snapshot of the reviewer network.

XII. Results and Discussion

A. Experimental Setup

All experiments were run on a standard laptop with Python 3.10. The dataset was split 70/30 (training/test), with stratification on the fake label. The primary metric is the F1-score on the fake class.

B. Model Performance

Table I summarises the F1-scores on the test set (example numbers; replace with your actual results).

TABLE I
F1-score for fake class (test set)

Model	F1-score (fake)
Logistic Regression	0.00
SVM	0.00
Random Forest	0.00
Gradient Boosting	0.25

These results show Gradient Boosting achieving a non-zero F1 on the difficult, imbalanced fake class. Use them as a starting point for future improvements.

C. Network Insights

SNA visualisations highlight reviewers with unusually high degree or clustered behaviour. These reviewer hubs often deserve manual inspection and can be used as interpretability cues for flagged reviews.

XIII. Conclusion and Future Work

A. Conclusion

We proposed a compact hybrid pipeline that combines SNA, TF–IDF, and behavioural features to detect fake reviews. Gradient Boosting offered the best results on the minority fake class in our experiments, and the pipeline was packaged into a lightweight demo.

B. Future Work

Possible extensions include: leftmargin=*

- richer network metrics (betweenness, community detection);
- contextual text embeddings (BERT/RoBERTa) instead of TF–IDF;
- temporal modelling to capture bursts and time-based attacks;
- robust anti-adversarial measures (defence against AI-generated or poisoned data).

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